

■ Machine Learning Formula Handbook (Running PDF)

■■ Date: 08/12/2025 – Linear Regression & Gradient Descent

Regression equation, MSE, gradient descent updates.

■■ Date: 09/12/2025 – Tree, SVM, Logistic, Scaling

Entropy, Gini, Sigmoid, MinMax scaling.

■■ Date: 10/12/2025 – Naive Bayes

Bayes theorem, Gaussian distribution.

■■ Date: 13/12/2025 – Kernel Functions

Linear, Polynomial, RBF, Sigmoid kernels.

■■ Date: 13/12/2025 – Regularization & Lasso

L1 and L2 regularization objective functions.

■■ Date: 13/12/2025 – KNN

Distance metrics.

■■ Date: 13/12/2025 – Classification Metrics

Precision, Recall, F1-score formulas.

■■ Date: 13/12/2025 – PCA

Covariance matrix, eigenvalues, eigenvectors.

■■ Date: 13/12/2025 – Bias & Variance

Trade-off summary.

■■ Date: 13/12/2025 – Bagging

Bootstrap aggregation logic.

■■ Date: 13/12/2025 – Image Preprocessing (OpenCV)

Normalization and resizing.

■■ Date: 13/12/2025 – Wavelet Transform

Time-frequency decomposition concepts.

■■ Date: 13/12/2025 – Feature Engineering

Scaling, encoding, transformations.

■■ Date: 13/12/2025 – Outlier Detection

Z-score and IQR detection concepts.

■■ Date: 13/12/2025 – Z-Score & Standard Deviation

Standard deviation, mean, z-score thresholds.

■■ Date: 13/12/2025 – Bell Curve (Normal Distribution)

Gaussian distribution properties and probability density.

■■ Date: 13/12/2025 – IQR (Interquartile Range)

Q1, Q3, IQR-based outlier thresholds.